

Christopher Nellis

Work Experience

Duke Energy

Principal Software Engineer

Sep 2025 — Present

- Founding member of Software Engineering Council, a cross-functional group focused on authoring, reviewing, and enforcing software engineering standards across the enterprise
- Served as tech lead for the implementation of Autho for customer authentication across all Duke Energy customer-facing applications, collaborating with CyberSecurity, legal, and various platform teams
- Designed, developed, and implemented Autho Actions to enhance and secure customer authentication flows according to requirements
- Served as subject matter expert for additional Autho tenants across the enterprise
- Developed Aurora Postgres database standards across the the enterprise including credential rotation, audit logging, and upgrade procedures
- Implemented SSO for Apple's AppStoreConnect platform across Duke Energy
- Consulted with CyberSecurity teams to disposition CodeQL findings, setting remediation plans or flagging false positives as necessary
- Partnered with Akamai firewall team to configure integrations for multiple teams, collaborated during during incidents, evaluated proposed firewall changes, and assisted in analyzing network traffic

Lead Software Engineer

Jun 2021 — Sep 2025

- Pioneered usage of ECS and Fargate at the company, becoming the first team to use those technologies
- Established patterns and best practices for deploying to ECR
- Migrated the customer mobile app's on-prem SQL Server Database to Aurora Postgres
- Architected system to send push notifications within AWS, integrating with Firebase Cloud Messaging
- Regularly communicated with product owners, non-technical stakeholders, peers, and developers to ensure alignment in technical requirements and overall behavior of the system

Senior Software Engineer

Sep 2019 — Jun 2021

- Led backend development throughout SAP Billing System consolidation
- Mentored new backend developers on the team
- Advocated for test-driven development and unit testing across the backend platform
- Navigated the migration of runtime environments from Pivotal Cloud Foundry to Nirmata

Software Engineer

Oct 2017 — Sep 2019

- Developed initial feature set of the customer mobile app
- Coordinated deployment across multiple teams for initial customer mobile app release
- Instituted coding challenges for hiring processes
- Built customer recommendation engine API layer with Spring

Revature

Software Developer

Aug 2016 — Oct 2017

- Participated on development teams for internal projects
- Presented technical solutions to colleges and trainers

UNC Office of Institutional Research and Assessment

Student Worker

May 2015 — Aug 2016

- Built tool to upload accreditation evidence and track approval status
- Managed HTML copy of accreditation report
- Created tool to translate addresses into latitude and longitude coordinates using Google Maps API
- Wrote PowerShell script to rename multiple files at once

Skills

- **Spring Boot Development:** Springcloud for AWS Integrations, Spring Web, Spring Security, Spring Data, Springdoc OpenAPI, Hibernate, Lombok
- **AWS Services:** IAM, Elastic Container Service, Fargate, Elastic Container Registry, API Gateway, Lambda, Cloudwatch, ElastiCache, Aurora, DynamoDB, Secrets Manager, Route53, Simple Queue Service, Simple Storage Service, Elastic Load Balancer, Database Migration Service
- **Enterprise Tools:** Terraform, Akamai, JIRA, GitHub, GitHub Actions for CI/CD, CodeQL, Dependabot, Flyway, Dynatrace, Hashicorp Vault

Education

University of North Carolina - Chapel Hill

BS in Information Science

May 2016

- Programming in Python
- Relational Databases
- Web Development
- Systems Analysis and Design
- Text Mining